Investigations on the at-issue status of viewpoint gestures
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Abstract

Most accounts on the semantic contribution of *co-speech gestures* posit that they contribute *not-at-issue* meaning by default (e.g., Ebert & Ebert 2014). This claim has been experimentally validated by Ebert, Ebert & Hörnig (2020) for gestures conveying information about an object's size or shape. In this article, the findings from two rating studies are reported investigating the at-issue status of *character* and *observer viewpoint gestures* (CVGs and OVGs, respectively, McNeill 1992). It is questionable whether the findings of Ebert, Ebert & Hörnig's (2020) study hold equally strong for CVGs and OVGs, as especially the former gesture type differs size from the gestures used in their study. The results of the studies suggest that while both gesture types contribute not-at-issue meaning by default, CVGs can shift more toward the at-issue dimension than OVGs if they provide information relevant to the *question under discussion*, in line with a gradient approach to at-issueness (Barnes & Ebert 2023).
Keywords: At-issueness, gesture semantics, perspective taking, viewpoint gestures, question under discussion

1 Introduction

Recently, there has been an increasing interest in research on the *at-issue status* of *iconic enrichments* (e.g., Ebert & Ebert 2014; Schlenker 2018a,b; Esipova 2019; Barnes et al. 2022) and thus also on the at-issue status of *iconic gestures*. Iconic gestures are gestures that resemble what they describe in certain (contextually relevant) aspects. Consider (1)[1]:

(1) There is <u>a swimming pool</u> in the garden. + **oval gesture**

The example shows that the oval gesture that is aligned with the indefinite NP *a swimming pool* adds information to the utterance, that is, it provides the information that the pool has an oval shape. However, it has been argued that this information is contributed in a different meaning dimension than the asserted content: the *not-at-issue* dimension (cf. Ebert & Ebert 2014; Schlenker 2018a). The multidimensionality of meaning contributions can be traced back to Potts' (2005) work on appositives.

(2) My friend Sophie, a classical violinist, performed a piece by Mozart.

(Syrett & Koev 2015: 525)

The sentence in (2) expresses two propositions. First, the at-issue proposition of the asserted part of the sentence, i.e., the main clause proposition that the speaker's friend Sophie performed a piece by Mozart. The second proposition is the not-at-issue proposition expressed by the appositive that Sophie is a classical violinist.

The two types of propositions differ in their semantic behavior as well as their discourse behavior. While at-issue content is on the table for discussion (Farkas & Bruce 2010), not-at-issue content is an aside that the speaker does not wish to discuss. Due to this property of not-at-issue content, it follows that it cannot be directly denied in a discourse:

[1] Underlining is used in examples to indicate gesture-speech alignment.

(3) A: My friend Sophie, a classical violinist, performed a piece by Mozart.
B: That's not true! # Sophie isn't a classical violinist.

Ebert & Ebert (2014) scrutinize the semantic contribution of co-speech gestures and note that they behave similarly to appositives. Therefore, they argue, the sentence in (1) also expresses two propositions, namely the at-issue proposition that there is a swimming pool in the garden and the not-at-issue proposition that this swimming pool has an oval shape. This claim has been verified experimentally by Ebert, Ebert & Hörnig (2020) for co-speech gestures providing information about an object's shape, i.e., gestures that do not encode viewpoint.

An additional dimension of gestures is how *viewpoint* is encoded in iconic gestures (McNeill 1992). Specifically, the most important distinction in this respect is the one between *character* and *observer viewpoint gestures* (CVGs and OVGs, respectively). While CVGs depict an event from a first-person and thus an internal perspective, OVGs depict events from a more external, i.e., a third-person perspective. Imagine you want to describe to someone an event where some person ran, and you also want to use an iconic gesture when describing this event. This can be done either by means of a CVG where an internal perspective is adopted and the running is thus depicted with the whole body, or alternatively by means of an OVG where the speaker uses their index and middle finger to represent the running person's legs. The iconic mapping is thus more abstract for OVGs than for CVGs. It has been shown that, among other things, CVGs and OVGs differ with respect to their size and their informativity (Beattie & Shovelton 2002).

The present work addresses two related research questions:

(4) 1. What is the at-issue status of CVGs and OVGs?
2. Does the at-issue status between CVGs and OVGs differ?

It might be surprising why the at-issue status of viewpoint gestures should be investigated at all, as there is already experimental evidence that co-speech gestures contribute not-at-issue meaning by default (Ebert, Ebert & Hörnig 2020). However, as noted above, especially for CVGs, it is unclear whether these experimental findings can be straightforwardly applied to them because they differ in terms of size from the gestures used in Ebert, Ebert & Hörnig's (2020) study. For the experiments reported in the present work, it was hypothesized that while both types of viewpoint gestures contribute not-at-issue meaning by default, CVGs are somewhat more at-issue than OVGs due to the aforementioned differences in size and informativity. This hypothesis is in line with a gradient approach to at-issueness (Barnes & Ebert 2023).

In order to tackle the research questions stated in (4), two experimental studies were conducted. The first one adapts Ebert, Ebert & Hörnig's (2020) paradigm. The results of the first experiment seem to corroborate the hypothesis that although CVGs and OVGs both contribute not-at-issue information by default, CVGs are more at-issue than OVGs. However, there might have been a bias in the design of the first experiment. Therefore, a second experiment was conducted. The results this time only confirmed the hypothesis partly: while again, the data clearly suggests that CVGs and OVGs contribute not-at-issue meaning by default, no difference in their at-issue status could be observed in the second experiment. Taking the results of both studies together, it is concluded that CVGs and OVGs are both equally not-at-issue by default. If, however, they provide information relevant to the *question under discussion*, CVGs can shift more toward the at-issue dimension than OVGs, in line with the gradient account to at-issueness proposed by Barnes & Ebert (2023). It remains an open question for now why CVGs can become more at-issue than OVGs.

The article is structured as follows: Section 2 provides the relevant background on at-issueness. In Section 3, Ebert & Ebert's (2014) semantic account of co-speech gestures is first discussed. Then, the findings of Ebert, Ebert & Hörnig (2020) are briefly summarized. They suggest that at-issueness is better captured as a gradient notion than a binary one. Thus, Barnes & Ebert's (2023) proposal to model gradient at-issueness is discussed in Section 3.3. Then, in Sections 4 and 5, the two experiments are reported. The paper concludes with a general discussion in Section 6.

2 (Not-)at-issueness

Traditionally, the at-issue part of an utterance is conceptualized as the primary meaning conveyed by that utterance, or, in Potts' (2005: 7) words, the part that "carr[ies] the main themes of a discourse." Moreover, the at-issue part of an utterance is what the interlocutors are expecting to negotiate about before it can enter the common ground, i.e., before it is accepted as shared knowledge (Potts 2007). Not-at-issue content, by contrast, is seen as an aside the speaker does not want to discuss. Therefore, it has been claimed that at-issue content is a proposal for a common ground update, whereas not-at-issue content is silently imposed on the common ground (e.g., AnderBois, Brasoveanu & Henderson 2015; Murray 2017). As will be shown below in more detail, at-issue and not-at-issue content differ in their discourse behavior, one of the most important differences between the two types of content being that not-at-issue content cannot be directly denied in discourse (cf. (11)). Moreover, not-at-issue content is projective, meaning that the speaker seems to be committed to it even if it is in the scope of entailment-canceling operators such as negation (e.g., Chierchia & McConnell-Ginet 1990). Consider the following example (adapted from Syrett & Koev 2015):

(5) It is not the case that my friend Sophie, a classical violinist, performed a piece by Mozart.
 a. She performed a piece by Bach.
 b. #She plays the trumpet.

A continuation as in (5b) contradicting the not-at-issue proposition expressed by the appositive, i.e., that Sophie is a classical violinist, is infelicitous. Thus, it is not targeted by the sentential negation. By contrast, the continuation in (5a) that contradicts the at-issue main clause proposition is felicitous. Projectivity can be seen as one of the defining characteristics of not-at-issue content (e.g., Tonhauser, Beaver & Degen 2018).

Koev (2018) notes that previous research on at-issueness has essentially resulted in three different notions of it: i) Q(uestion)-at-issueness, P(roposal)-at-issueness, and iii) C(oherence)-at-issueness. Note, however, that many theories modeling at-issueness cannot be strictly assigned to one of these notions. Since for the purposes of the present paper, a mix of Q- and P-at-issueness will be assumed, only these two notions will be discussed in detail below.

Q-at-issueness takes topicality to be one of the organizing principles of the discourse (Koev 2018). The core assumption in these theories is that discourse is structured according to potentially covert *questions under discussion* (QUDs). These QUDs are partially ordered by the informativity of their answers. When uttering a sentence, multiple propositions are often expressed. However, not all of them are (equally) relevant to the current QUD. A proposition conveyed in an utterance can be called at-issue when it informs the current QUD (Simons et al. 2010), leading to the following definition of Q-at-issueness:

(6) A proposition p is Q-at-issue relative to a QUD Q in a context c iff p is relevant to Q in c and p is appropriately conventionally marked relative to Q.

(Koev 2018: 3)

The first condition in (6) relates to the Gricean maxim of relevance (Grice 1975), the main idea being that a proposition has to provide information to interlocutors that helps them decide the relevant alternatives raised by the QUD in order for it to be Q-at-issue. Relevance is defined in terms of (7).

(7) A proposition p is RELEVANT to a QUD Q and a context c iff for some $q \in Q : Pr_c(q) \neq Pr_c(q|p)$.

(Koev 2018: 3)

Pr_c in (7) is the probability relative to a context c and $Pr(q|p)$ is the conditional probability of q given p.

The most straightforward way of applying (7) in discourse is when a response to a QUD selects one alternative and assigns to it the truth value 1 and 0 to all other alternatives. This, however, is not the typical way discourse is structured. Rather, only partial answers to the QUD are given. Consider (8).

(8) Q: What did Elijah eat?
A: He didn't eat salmon(, that's all I know).

The answer in (8) does not provide a full answer to the question. Instead, it gives a partial answer about what Elijah did not eat, thus reducing the set of possible alternatives, but not narrowing it down to a single alternative. A reply like *He ate falafel*, by contrast, would do so. Hence, partial answers are informative but not as informative as full answers.

The second part of the definition provided in (6), namely the appropriate conventional marking of the at-issue content relative to the QUD, is illustrated by (9). The example shows that an answer to a QUD can be infelicitous even if it contains the relevant information to answer it, but is not appropriately marked.

(9) Q: What happened after the spacecraft sent back the first pictures?
A: #There was water on Mars, NASA announced.

(Koev 2018: 4)

The answer in (9) is infelicitous because the relevant information is given as a parenthetical. Following Beaver et al. (2017), "appropriate conventional marking [is] the restriction that only content whose triggering expression obligatorily contributes to the truth conditions of the host clause can be Q-at issue" (Koev 2018: 4). In other words, this means that projective inferences, such as presuppositions and conventional implicatures and therefore also appositive relative clauses, cannot be Q-at-issue. Summing up, Q-at-issueness is strongly dependent on the topic of the current discourse, which can be neatly modeled via the assumption of potentially covert QUDs.

A further notion of at-issueness that has been discussed in the literature relates to the so-called *common ground* or *context set* (Stalnaker 1978). The common ground is seen as the set of propositions which are believed by all interlocutors in a discourse. The context set is built by intersecting over the common ground, which yields a set of possible worlds, namely the set of exactly those worlds that are contained in every proposition in the common ground. If an interlocutor makes an assertion, this is seen as a proposal to update the common ground with that proposition. In terms of the context set, the assertion is a proposal to reduce the context set to those worlds where the assertion's content is true. If this assertion is accepted by all interlocutors, the proposition is added to the common ground, or, in terms of the context set, it is reduced to those worlds where the content of the assertion holds. This is the most important function of assertions (Stalnaker 1978).

How do context set and assertions relate to at-issueness? The first relation between them and at-issueness is based on proposalhood (Koev 2018). If a proposed assertion is accepted by all interlocutors, the context set is reduced, as it eliminates exactly those worlds from the context set which are not contained in the proposition expressed by the sentence uttered. The second feature, although not fleshed out in Stalnaker's (1978) pioneering work, is that an assertion, if it is sincere, comes with speaker commitments (cf. Farkas & Bruce 2010, among others). Based on this background, the at-issue part of an utterance is seen as the proposal which is introduced when asserting a declarative sentence (Potts, 2005; AnderBois, Brasoveanu & Henderson 2015, among others). Since this notion is closely tied to the proposal a speaker makes when making an assertion, Koev (2018) dubs this notion Proposal-at-issueness, abbreviated as P-at-issueness:

(10) A proposition p is P-at-issue in a context c iff p is a proposal in c and p has not been accepted or rejected in c.

(Koev 2018: 5)

Crucially, only content that is not yet mutually accepted by all interlocutors can be P-at-issue. If it has entered the context set, it cannot be at-issue any longer, as it fails to meet the defining criterion in (10). Furthermore, note that, according to Koev (2018), theories of P-at-issueness often also incorporate ideas from the aforementioned notion of Q-at-issueness. Therefore, as stated above, a clear-cut distinction between these notions is not always possible.

There are several tests available in order to determine the P-at-issue status of an asserted proposition. A frequently used diagnostic is the so-called *assent/dissent test*. This test checks whether an expression can be directly denied in discourse. If it can be denied, it is P-at-issue because it is still debatable, meaning that it has not entered the common ground yet. If this is not the case, it is not P-at-issue. An example based on Amaral, Roberts & Smith (2007) is given in (11).

(11) A: Edna, a fearless leader, started the descent.
B: That's not true—Edna has not started the descent.
B': #That's not true—Edna is not a fearless leader.

(Koev 2018: 6)

This example shows that only the main clause of A's utterance is P-at-issue. The nominal appositive *a fearless leader*, by contrast, is not P-at-issue as it cannot be directly dissented with.

So far, a test to diagnose whether an expression is P-at-issue has been introduced. Consequently, it only allows us to test for not-P-at-issueness indirectly. However, there is a test to directly identify content which is not P-at-issue, originally used to identify presuppositions (cf. Shanon 1976; von Fintel 2004). An example testing a presupposition is given in (12) (taken from von Fintel 2004: 3).

(12) A: The mathematician who proved Goldbach's Conjecture is a woman.
B: Hey, wait a minute. I had no idea that someone proved Goldbach's Conjecture.
B': #Hey, wait a minute. I had no idea that that was a woman.

The so-called *Hey, wait a minute!* test supposedly works only when targeting the presuppositional content in A's utterance in (12). Under the assumption that all projective content, such as presuppositions and conventional implicatures (e.g., appositives), contributes not-at-issue information by default (Beaver 2001; Simons et al. 2010, among others), this test can be extended to not-P-at-issue content in general.

It has been shown so far that there are several notions of at-issueness in linguistics. Although all three notions overlap to a certain extent, their core assumptions are conceptualized with respect to different theoretical approaches. While Q-at-issueness is strongly related to the QUD, the core assumptions of P-at-issueness, by contrast, are based within Stalnaker's (1978) theory of common ground. In this paper, a mix between P- and Q-at-issueness will be assumed and, for reasons of simplicity, from now on only referred to as (not-)at-issueness. Specifically, not-at-issue propositions are not available for discussion and thus part of the context set (relating to P-at-issueness) while it is at the same time assumed that only those propositions can be at-issue in the first place that give at least a partial answer to a (potentially) covert QUD, which relates to Q-at-issueness.

3 Co-speech gestures

3.1 How do gestures contribute meaning?

The pioneering approach to systematically model the semantic contribution of *iconic gestures* and *pointing gestures* was put forth by Ebert & Ebert (2014).[2] Under their approach, iconic and pointing gestures receive a uniform formal treatment. Therefore, it is irrelevant whether examples with pointing gestures or with iconic gestures are used for illustration.

Before providing a detailed summary of the formal apparatus proposed by Ebert & Ebert (2014), it is worth noting some general properties of co-speech gestures. They are communicative movements of hands, arms, and also other body parts conveying, among other things, thoughts, emotions, and intentions. Iconic gestures are not conventionalized, meaning that there is a multitude of iconic gestures available to speakers to express the same concept. Gestures contribute meaning to an utterance. Speech and gesture work together to express one thought, but in different modalities (McNeill 1992).

Different types of gestures have been identified in gesture research, including iconic gestures and pointing gestures. Pointing gestures are often imprecise and, as a result, ambiguous. For instance, when a speaker points to a chair, it may be unclear whether they are referring to the entire chair or only specific parts of it. Additionally, pointing gestures are deictic, meaning they depend on the context for interpretation.

In contrast, iconic gestures resemble specific aspects of an entity or action. For example, the *rectangular* gesture in (13) illustrates a similarity to squareness or a rectangular object. Such gestures inherently convey visual or conceptual resemblance.

(13) I saw <u>a window</u> in the living room. + **rectangular co-speech gesture**

A further important finding of previous gesture research is that gesture and speech are temporally aligned (Kendon 1980; McNeill 1992). A gesture can be divided into several so-called *gesture phases*, which in turn combine to form a *gesture phrase*. A gesture phrase consists of three phases: i) the preparation phase, where the speaker brings their hands, arms, and possibly their whole body into a position where they can perform the gesture, ii) the stroke, the main part of the gesture where the main movement of the hands/body takes place, and finally iii) the retraction phase, in which hands and arms are brought back into resting position. Experimental

[2]The term *gesture* is used sloppily throughout this paper. If not specified otherwise, the term *gesture* will be used to mean co-speech gesture.

investigations show that the stroke, for example, is temporally aligned to the nuclear accent of a word. It cannot occur at any later point (McNeill, 1992; Ebert, Evert & Wilmes 2011, among others). This constitutes a further piece of evidence that gesture and speech are intertwined systematically.

Co-speech gestures contribute not-at-issue meaning by default (Ebert & Ebert 2014; Schlenker 2018a; Ebert, Ebert & Hörnig 2020). Consider example (13) again. As pointed out in the previous section, a defining property of not-at-issueness as it is understood in this paper is that not-at-issue content cannot be directly denied in discourse (cf. (11)). If a speaker utters the sentence in (13) and the addressee wants to object against the window's shape, which is provided via the gesture, they cannot directly do this (cf. (14)). What they have to do instead is to use one of the strategies employed in (14b) and (14c).

(14) a. #That's not true. The window is actually of a round shape.
b. Hey, wait a minute! The window is not rectangular, it's actually round.
c. Yes, but the window is not rectangular, it is round.

The *Hey, wait a minute!* test as shown in (14b) shows that a discourse-interrupting element has to be used to target the content of the gesture (cf. von Fintel 2004). A further strategy is to agree with the general statement, as in (14c), and to then challenge the gestural content (cf. Tonhauser's, 2012 diagnostic #1c). The claim that co-speech gestures contribute not-at-issue meaning by default has been experimentally validated by Ebert, Ebert & Hörnig (2020) whose findings will be summarized in Section 3.2 of the present work.

Gestural content also projects through negation and other semantic operators (Ebert & Ebert 2014) which is illustrated by (15).

(15) a. I did not see <u>a window</u> in the living room. #I only saw a round one. **+ rectangular co-speech gesture**
b. I did not see a rectangular window in the living room. I only saw a round one.

The continuation in (15a) is infelicitous, showing that the gestural content projects through the negation. If its content did not project through the negation, the continuation would be expected to be felicitous. (15b) further strengthens this. In this example, the shape information of the window is contributed via speech, i.e., via the adjective *rectangular*. Interestingly, co-speech gestures pattern with appositives in this respect. Ebert & Ebert (2014) therefore analyze them as conventional implicatures in the sense of Potts (2005).

Sentences accompanied by a co-speech gesture thus contribute meaning in two dimensions, just like appositives (cf. Potts 2005): they make an at-issue proposal and a not-at-issue imposition (AnderBois, Brasoveanu & Henderson 2015). For (13), this is paraphrased in (16).

(16) a. at-issue: The speaker saw a window in the living room.
b. not-at-issue: The window is rectangular.

More generally, the at-issue dimension of a multimodal utterance consists of the spoken utterance's semantic content. The not-at-issue dimension, by contrast, consists of the gesture's semantic content.

In Ebert & Ebert's (2014) analysis, pointing and iconic gestures both refer to an individual. This individual is abstract for iconic gestures. It must at least carry the crucial comparison features (cf. Umbach & Gust 2014) which are contextually specified. Pointing gestures refer directly to the individual which is pointed at. *Deferred reference* for pointing gestures, however, is also possible (Nunberg 1993), meaning that the intended referent does not have to correspond to the pointing referent. For example, if a speaker points to a picture of someone, they are referring to the person shown in the picture, not the picture itself.

Furthermore, the temporal alignment of gesture and speech is a decisive feature specifying the gesture's meaning contribution in Ebert & Ebert's (2014) approach. Beside its core meaning, the production of a gesture yields a so-called *constructional meaning* (McNeill 1992, among others) which is dependent on the verbal expression it co-occurs with. Specifically, they argue that the alignment of either an iconic or a pointing gesture to an indefinite article yields an interpretation under which the gestural referent g is similar to the verbal referent (cf. Umbach & Gust 2014). When a gesture is aligned to a proper name or the definite article, this expresses identity between the gestural and the verbal referent. Finally, if the gesture is aligned with an NP, g exemplifies a verbal concept. These claims have been experimentally validated by Ebert, Pirillo & Walter (2022).

To formally model this, Ebert & Ebert (2014) adapt the unidimensional and dynamic system proposed by AnderBois, Brasoveanu & Henderson (2015). In the system, variables, such as x, y, and z, stand for individual concepts and are hence of the semantic type $\langle \mathrm{s},\ \mathrm{e} \rangle$. They are introduced by the dynamic existential quantifier [·]. In order to keep track of the different dimensions of meaning, the propositional variables p for the at-issue dimension and p^* for the not-at-issue dimension are introduced into the system (cf. AnderBois, Brasoveanu & Henderson 2015). The variable p introduces a proposal to update the common ground; p^*, by contrast, imposes its content on the common ground, thus relating to the notion of P-at-issueness discussed in Section 2 of this paper. The gestural referent g denotes a rigid designator R_g, meaning that in all worlds $w \in W$, it returns the same value, i.e., the intended gesture referent g. The performance of a speech-accompanying gesture, then, introduces a new discourse referent for R_g.

(17) $[\mathrm{z}] \wedge \mathrm{z} = \mathrm{R}_g$, where for all $\mathrm{w} \in \mathrm{W}$: $[\![\mathrm{R}_g(\mathrm{w})]\!] = \mathrm{g}$

Aligning the gesture with an indefinite article expresses a similarity relation (cf. Umbach & Gust 2014). The variables x and z stand for the referents introduced by the indefinite and the gestural referent, respectively.

(18) $\mathrm{SIM}_{p^*}(\mathrm{x},\ \mathrm{z})$

The similarity relation is relativized to the not-at-issue variable p^* and is true iff z is similar to x in the contextually relevant aspects.

The alignment of a gesture to a name or the definite article expresses identity between the definite referent x and the gestural referent z.

(19) $\mathrm{x} =_{p^*} \mathrm{z}$

It is again relativized to the not-at-issue propositional variable p^* and true just in case x and z designate the same object. Additionally, existence and uniqueness presuppositions are triggered by the definite determiner.

The temporal alignment of an NP and a gesture triggers an exemplification interpretation, that is, the property expressed by a noun N is true of the gestural referent z.

(20) $\mathrm{N}_{p^*}(\mathrm{z})$

This contribution is again relativized to p^* and therefore contributes not-at-issue meaning.

Consider the following multimodal utterance:

(21) My living room has <u>a window</u>. + **rectangular co-speech gesture**

at-issue: The speaker's living room has a window.
not-at-issue: The gesture referent is similar to this window, and the gesture referent itself is a window.

(cf. Ebert, Pirillo & Walter 2022: 70)

Ebert & Ebert (2014) formalize this sentence as in (22):

(22) $[x] \wedge x = my\_living\_room \wedge [y] \wedge window_p(y) \wedge [z] \wedge z = R_g \wedge SIM_{p^*}(y, z) \wedge window_{p^*}(z) \wedge has_p(x, y)$

(Ebert, Pirillo & Walter 2022: 70)

The formula in (22) introduces an individual concept x and is assigned the value of the speaker's living room. Then, a second individual concept y is introduced, which is ascribed the at-issue property of being a window. In a next step, a further individual concept z is introduced which is assigned the value of the rigid designator denoted by the gesture. The alignment of the gesture to the indefinite article triggers a not-at-issue similarity interpretation. Finally, the verb *has* makes an at-issue proposal that x has y.

Alignment of the gesture to the NP (cf. (23)) is formalized as in (24).

(23) My living room has a window. + **rectangular co-speech gesture**
at-issue: The speaker's living room has a window.
not-at-issue: The gesture referent is itself a window.

(cf. Ebert, Pirillo & Walter 2022: 70)

(24) $[x] \wedge x = my\_living\_room \wedge [y] \wedge window_p(y) \wedge [z] \wedge z = R_g \wedge window_{p^*}(z) \wedge has_p(x, y)$

The formula in itself is very similar to that in (22), the sole difference being that the SIM-predicate is missing since NP-alignment of the gesture only triggers an exemplification interpretation.

A calculation for an instance where the gesture temporally aligns with the definite determiner (cf. (25)) is given in (26).

(25) Peter saw the window. + **rectangular co-speech gesture**
presupposition: There is a unique contextually salient window.
at-issue: Peter saw the window.
not-at-issue: The gesture referent is the window, and the gesture referent is itself a window.

As already stated above, the definite determiner triggers an existence and a uniqueness presupposition.

(26) $[x] \wedge x = peter \wedge [y] \wedge window_p(y) \wedge [z] \wedge z = R_g \wedge y =_{p^*} z \wedge window_{p^*}(z) \wedge saw_p(x, y)$

This formula assigns the value $peter$ to the individual concept variable x. Next, it introduces a further variable y and assigns to it the at-issue property of being a window. Then, the variable z is introduced to which the value of the rigid designator R_g is assigned. In a next step, the not-at-issue identity relation is introduced, which states that relative to p^* worlds, y and z are identical. Afterwards the not-at-issue property of being a window is assigned to z. Finally, the at-issue contribution is made that x saw y.

Ebert & Ebert (2014) predict a further alignment pattern of gesture and speech, namely the alignment of a co-speech gesture to a demonstrative, such as the German similarity demonstrative *SO* ('so/like this/such', it is capitalized as it normally receives a focal accent).

(27) A: Peter hat <u>SO</u> ein Fenster gesehen. + **rectangular co-speech gesture**
'Peter has seen a window <u>like this</u>.'
B: That's not true! The window was actually round.

Interestingly, when the gesture is temporally aligned with *SO*, its content can be directly denied. Moreover, as (28) illustrates, the gestural content can be targeted by negation. This implies that it is not projective in this utterance.

(28) Peter hat nicht <u>SO</u> ein Fenster gesehen. Es war nämlich rund. + **rectangular co-speech gesture**
'Peter has not seen a window <u>like this</u>. Actually, it was round.'

It can be concluded from these two examples that co-speech gestures contribute at-issue information when they are aligned with a demonstrative. Based on this, Ebert & Ebert (2014) propose to analyze demonstratives as *dimension shifters*, i.e., they are seen as semantically vacuous expressions with the sole function of shifting the gesture that co-occurs with it to the at-issue dimension. In later work (cf. Ebert & Ebert 2021), the meaning contribution of *SO* is formally implemented as in (29).

(29) $[\![\Uparrow^{p}_{p^*} (\phi_p \wedge \psi_{p^*})]\!] \equiv [\![\phi_p \wedge \psi_p]\!]$

(Ebert & Ebert 2021: 18)

The subscript behind the operator $\Uparrow$ indicates from which dimension the content is shifted and the superscript shows to which dimension the content is shifted to. Hence, this operator takes an expression which is relativized to p^*, i.e., the not-at-issue dimension, and shifts it to the at-issue dimension, meaning that it is then relativized to the at-issue propositional variable p. In (29), this means that the operator takes the expression ψ, expressing a not-at-issue proposition, and shifts it to the at-issue dimension.

The formalization of the sentence containing the gesture in (27) is thus as follows:

(30) $\Uparrow^{p}_{p^*}$([x] $\wedge$ x = peter $\wedge$ [y] $\wedge$ window$_p$(y) $\wedge$ [z] $\wedge$ z = R$_g$ $\wedge$ SIM$_{p^*}$(y, z) $\wedge$ window$_{p^*}$(z) $\wedge$ saw$_p$(x, y)) $\equiv$
[x] $\wedge$ x = peter $\wedge$ [y] $\wedge$ window$_p$(y) $\wedge$ [z] $\wedge$ z = R$_g$ $\wedge$ SIM$_p$(y, z) $\wedge$ window$_{p^*}$(z) $\wedge$ saw$_p$(x, y)

It is important to note that in (30) only the SIM-predicate is shifted. The not-at-issue contribution that the gesture referent itself is a window is not shifted. The demonstrative *diese* (Engl. 'this') is treated as a shifted version of the definite determiner. Therefore, when aligning a co-speech gesture with *diese*, this triggers an interpretation as stated in (26), the only difference being that the relativized identity between y and z is shifted to the at-issue dimension, i.e., the identity statement is then relativized to p.

3.2 Experimental research on the at-issue status of co-speech gestures

In order to test for the hypotheses that i) co-speech gestures contribute not-at-issue meaning by default and ii) that a demonstrative like German *SO* (‘so/like this/such’) can shift a gesture’s meaning contribution to the at-issue dimension, Ebert, Ebert & Hörnig (2020) conducted two rating studies. The first experiment addressed the first hypothesis. Based on findings that false information in not-at-issue content does not affect the overall truth conditions of an utterance as strongly as false information stemming from at-issue content does (Kroll & Rysling 2019), Ebert, Ebert & Hörnig (2020) test the at-issue status of co-speech gestures by means of a sentence-picture matching task. In this task, participants had to rate how well an utterance which contained either a co-speech gesture giving information about an object’s shape or an adjective giving the same information as the gesture in the spoken modality matches a picture showing such an object. Crucially, the object shown in the picture could either match or not match the shape information provided by the gesture or the adjective, respectively. The study was thus of a 2x2 design with the factors Match (matching vs. mismatching picture) and Mode (gesture vs. adjective). An example of an experimental item can be found in (31).

(31) a. Auf diesem Bild ist eine Mauer mit <u>einem Fenster</u> zu sehen. **+ circular co-speech gesture**
‘In this picture you can see a wall with <u>a window</u>.’
b. Auf diesem Bild ist eine Mauer mit einem **runden** Fenster zu sehen.
‘In this picture you can see a wall with a round window.’

(Ebert, Ebert & Hörnig 2020: 173)

Based on the hypothesis that co-speech gestures contribute not-at-issue meaning by default and thus less severely affect the truth conditions, Ebert, Ebert & Hörnig (2020) predicted a higher mismatch effect, i.e., the rating difference between the matching and the mismatching condition, for the adjective control condition than for the gesture condition. The reasoning behind this prediction is as follows: the task requires participants to state whether (or not) they think the scenario described by the utterance matches the scenario shown in the picture. This means that, indirectly, they are asked to judge to which extent the utterance is true of the picture[3]. Based on the aforementioned finding that not-at-issue content influences the truth conditions of an utterance less severely than at-issue content does (Kroll & Rysling 2019), this should lead participants to judge the mismatching condition worse if the mismatching information is given by means of at-issue content (i.e., an adjective) compared to when it is expressed in the not-at-issue dimension (i.e., by a co-speech gesture). The results confirmed their hypothesis. Thus, co-speech gestures indeed contribute not-at-issue meaning by default.

In order to test for the second hypothesis that demonstratives are dimension shifters, Ebert, Ebert & Hörnig (2020) conducted a follow-up study with the same design as the first experiment. The only difference was that they added a condition to the Mode factor where a co-speech gesture was aligned with the German demonstrative *SO*:

(32) Auf diesem Bild ist eine Mauer mit <u>SO</u> einem Fenster zu sehen. **+ circular co-speech gesture**
‘In this picture you can see a wall with a window <u>like this</u>.’

(Ebert, Ebert & Hörnig 2020: 174)

[3]For a critical discussion of committing oneself to the tight connection between truth and acceptability adopted by Barnes & Ebert (2023) and also in this paper, see Koev (2023).

Since Ebert & Ebert's (2014) account predicts that demonstratives are dimension shifters, they hypothesized that the mismatch effect for the condition *SO* + gesture should be as high as in the adjectival control condition. Compared to the gesture condition, by contrast, there should be a significant rating difference. They thus predicted an interaction between MATCH and MODE again. Although the results confirm their prediction, their hypothesis is only partly borne out: the condition *SO* + gesture neither patterned with the gesture condition nor with the adjectival control condition. Instead, the ratings were in between the two other conditions. They therefore conclude that demonstratives do shift a gesture's meaning toward the at-issue dimension, but they do not become as at-issue as asserted material.

3.3 Gradient at-issueness

The findings of the experimental studies reported in Ebert, Ebert & Hörnig (2020) as well as similar findings on German adverbial ideophones (Barnes et al. 2022) pose challenges to traditional, that is binary, notions of at-issueness. Under the Q-at-issueness approach, an expression is at-issue if and only if it provides at least a partial answer to the QUD (cf. (6) in Section 2 of the present paper). However, the data obtained for *SO* + gesture, for example, indicate that *SO* shifts a gesture's meaning contribution toward the at-issue dimension, but it does not make its meaning contribution as at-issue as asserted content is. Instead, it can better be located somewhere between at-issue and not-at-issue. Similar observations have been made for projective content in general. Previous experimental findings suggest that projection is also gradient, as, for example, not all factive predicates (e.g., *know*) project equally strong and that projection variability is also subject to context (Xue & Onea 2011; Smith & Hall 2011; Tonhauser 2020; Degen & Tonhauser 2021, 2022). The view that projection is gradient has been most explicitly defended by Tonhauser, Beaver & Degen (2018: 499), where they posit the Gradient Projection Principle, stating that content expressed by some constituent which is embedded under an entailment-canceling operator projects to the extent that is not-at-issue.

The observations described above lead Barnes & Ebert (2023) to propose a gradient approach to at-issueness. They assume a QUD-based account to at-issueness and propose that every expression comes with an inherent at-issueness value that can be located on a scale. This value depends on numerous factors, such as the modality in which it occurs and its structural position. This also means that if an expression that normally is not used to address a QUD exceptionally informs the QUD, it can also shift toward the at-issue dimension. In their approach, an iconic co-speech gesture is assigned a low value of inherent at-issueness and thus, it is located at the not-at-issue end of the scale. By contrast, a gesture aligned with a demonstrative is ordered relatively high on the scale and is thus more at-issue. For the formal details, the reader is referred to their article. The scale Barnes & Ebert (2023) propose for co-speech gestures and sentence-medial adverbial ideophones is stated in (33). The symbol > means that the enrichment to the left is less at-issue than the enrichment to the right.

(33) iconic co-speech gestures > sentence-medial adverbial ideophones > DEM + iconic co-speech gestures > DEM + sentence-medial adverbial ideophones

(Barnes & Ebert 2023: 212)

The higher ranking of ideophones compared to co-speech gestures comes about due to the former being used in the same modality as asserted material in spoken language, while the latter occur in the visual modality, which is secondary in spoken languages (Barnes & Ebert 2023).

Figure 1: A skunk hopping across a room. Cartoon scene from Parrill (2010: 651)

(a) CVG used to depict the skunk in Figure 1 (Figure 3 in Parrill 2010: 652)

(b) OVG used to depict the skunk in Figure 1 (Figure 2 in Parrill 2010: 651)

Figure 2: Examples for a CVG and an OVG to depict the event shown in Figure 1

3.4 Viewpoint in gesture

While investigations on perspective-taking in spoken language have been of interest in linguistics for decades (see Hinterwimmer, 2024 for a recent overview on perspective in narrative texts), the investigation of visual means of perspective-taking has remained largely unexplored. McNeill (1992) notes, however, that perspective can also be encoded in gestures. He distinguishes *character viewpoint gestures* (CVGs), *observer viewpoint gestures* (OVGs), and *dual viewpoint gestures.* Properties of the former two are discussed in this section.

CVGs depict an event from an internal, first-person perspective. Therefore, the whole body is normally involved in the production of the gesture. By contrast, OVGs depict an event from a more external, i.e., third-person perspective and thus normally only hands and arms are involved when producing the gesture (McNeill 1992). Moreover, CVGs have been argued to be more informative than OVGs (Beattie & Shovelton 2002). There are additionally language-specific preferences concerning the preferred perspective encoded in a gesture (Özyürek et al. 2005; Stec 2012). This preference is so strong that already intermediate L2 learners adopt that language-specific strategy (Brown 2008).

Concerning language-internal factors determining the choice of the gestural perspective (henceforth *viewpoint* following McNeill's 1992 terminology), McNeill (1992) has noted that CVGs tend to co-occur with transitive utterances more often than with intransitive utterances. For discourse or event structure, CVGs have been argued to occur more frequently than OVGs with discourse-central events compared to discourse-peripheral events (McNeill 1992). These observations were based on a very small data set, however. Therefore, Parrill (2010) conducted a more comprehensive study aiming to investigate language-internal as well as discourse factors affecting gestural viewpoint choice.

In the study, participants watched short cartoon clips. They then had to explain the contents of the clips to a friend they brought to the study who had not seen them. These explanations were then videotaped and annotated. The results show that CVGs co-occur with transitive utterances more frequently than with intransitive utterances. For OVGs, the reverse pattern is observed: they co-occur with intransitive utterances more frequently than with transitive utterances, thus confirming McNeill's (1992) observation.

The hypothesis that CVGs occur more frequently with discourse-central events than with discourse-peripheral events is not borne out. Rather, both types of viewpoint gestures occur more frequently with discourse-central events than with discourse-peripheral events. A comparison between CVGs and OVGs did not yield a significant difference (Parrill 2010). The same pattern emerges when considering not the discourse-centrality of the event the gesture relates to, but the centrality of the utterance in the narration.

Event structure, however, has an effect on gestural viewpoint choice (Parrill 2010), since there were some events that evoked CVGs exclusively while others exclusively evoked OVGs. Finally, there were also some events that evoked CVGs and OVGs with approximately the same frequency.

CVG only events included those events where some form of handling, i.e., a character using their hands to accomplish a task, was involved. An example event was a character riding the bus, holding and reading a newspaper. It has to be noted here that this event would typically be described with a transitive utterance. However, as Parrill (2010) argues, transitivity might not be the driving factor here. Instead, exclusively CVGs were used because there is no apparent OVG alternative that is equally suitable to describe this type of event. The 18 CVG only events were grouped into three main categories: i) handling (n = 10), ii) emotional state or affect (such as laughter, n = 5), and iii) torso use (e.g., shrugging, n = 3). A fourth category was introduced including those events of the 18 CVG only events where no OVG was available to adequately describe that event (n = 3). This shows that the majority of events were such that there would have been a plausible OVG alternative, yet narrators decided to use a CVG instead.

All events involving some form of trajectory evoked OVGs exclusively. An example of this was an event where a character ran outside of a baseball stadium. Although trajectory-involving events are normally intransitive as they are of the structure subject-verb-prepositional phrase, a better explanation for this finding—according to Parrill (2010)—is that trajectory can simply be gestured more easily with an OVG than with a CVG. However, for some of the OVG only events, a CVG would have been in principle available, as, for example, an event of a skunk hopping across the room similar to that shown in Figure 1.

For those events evoking CVGs and OVGs with roughly the same frequency, discourse factors, such as the narrator's focus, seem to determine the choice of gestural viewpoint rather than the event structure itself. Interestingly, one of these events was very similar to an OVG only event, namely the event of the hopping skunk mentioned above. The only difference was that the event evoking CVGs and OVGs with the same frequency was that the skunk also chased a cat in the CVG and OVG evoking event. However, narrative structure seems to be the driving factor here (Parrill 2010). Concretely, the event evoking both types of viewpoint gestures was when the funny hopping of the skunk was introduced for the first time. Therefore, narrators had to choose whether they wanted to focus on the manner in which the hopping took place or on the trajectory of the hopping event. This resulted in a CVG or an OVG being produced, respectively. The OVG only event, by contrast, was introduced after that event. Therefore, the skunk's marked manner of hopping was already established, and therefore the trajectory played a more important role, thus evoking OVGs exclusively at that later point in the discourse. A final remark on this is that discourse centrality does not seem to be the driving factor for events evoking CVGs and OVGs with the same frequency, since not all discourse central events showed this variability (Parrill 2010).

In sum, the choice of gestural viewpoint is dependent on many different factors: while there are factors such as transitivity that affect this choice, discourse and event structure also play a crucial role.

Before turning to Experiment 1, let's summarize what has been discussed so far in order to formulate hypotheses concerning differences in the at-issue status between CVGs and OVGs. Section 2 has laid out the theoretical background on at-issueness. The key properties of not-at-issue content are i) that it projects through negation and other entailment-canceling operators, ii) that the proposition expressed by not-at-issue content is an imposed common ground update rather than a proposal to update the common ground (AnderBois, Brasoveanu & Henderson 2015), iii) it normally does not address the QUD (Simons et al. 2010), and iv) that it cannot be directly denied in discourse (von Fintel 2004). These properties have been shown to hold for co-speech gestures in Section 3.1. Their not-at-issue status has been experimentally verified by Ebert, Ebert & Hörnig (2020) (cf. Section 3.2). Their findings lead to Barnes & Ebert's (2023) proposal to capture at-issueness as a gradient notion (Section 3.3). Crucially, in their account, an expression can be shifted toward the at-issue dimension if it informs the QUD. In Section 3.4 iconic gestures encoding viewpoint, i.e., CVGs and OVGs, were discussed. Normally, a CVG is produced with the whole body and depicts an event from a first-person perspective, whereas only hands and arms are involved when producing an OVG, which depict an event from a third-person perspective. CVGs are thus larger in size than OVGs. CVGs have been shown to be more informative than OVGs (Beattie & Shovelton 2002) and there are language-specific preferences for which viewpoint is typically encoded in gesture (Özyürek et al. 2005). In addition, linguistic and event structure have been shown to affect gestural viewpoint, too (Parrill 2010). Based on these peculiarities of viewpoint gestures, it is questionable whether Ebert, Ebert & Hörnig's (2020) findings also hold for them. Due to the differences between CVGs and OVGs, especially those concerning their size and informativity, it is hypothesized for the experiments reported in Sections 4 and 5 that although CVGs and OVGs both contribute not-at-issue meaning by default, the former gesture type is still more at-issue than the latter gesture type. Thus, the gradient approach to at-issueness proposed by Barnes & Ebert (2023) is adopted in this paper.

4 Experiment 1

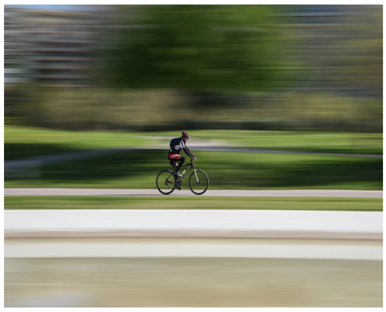
4.1 Method

4.1.1 Participants

Participants (n = 60) were recruited via Prolific[4]. They ranged from 18–70 years of age (M = 32.68, SD = 12.62). All of them were native speakers of German and naive with respect to the research question. They were compensated with £3.20 for participation.

[4] www.prolific.com

(a) Matching picture



(b) Mismatching picture

Figure 3: The matching and the mismatching picture for the experimental item in (34)

4.1.2 Materials

Eighteen experimental items were created for the experiment. Each item consisted of two sentences: the first sentence introduced an event or contained a general statement by the speaker, and the second sentence further described this event or general statement. The second sentence contained a word, usually a verb, which was either accompanied by a CVG or an OVG. Alternatively, a phrase paraphrasing the gesture information preceded or followed this word (factor Mode: CVG vs. OVG vs. Verbal). The gestures conveyed the same meaning, only differing in the perspectival information they encoded. Moreover, all experimental items were written from a first-person perspective. An experimental item is given in (34).

(34) a. Letzten Mittwoch hatte ich den ganzen Tag Termine überall in der Stadt. Nachdem einer der Termine länger dauerte als gedacht, musste ich mich richtig beeilen.
CVG: Speaker moves their arms and legs to demonstrate a running event.
OVG: Speaker uses two fingers of their right hand to depict a running event; each finger represents a leg.

b. Letzten Mittwoch hatte ich den ganzen Tag Termine überall in der Stadt. Nachdem einer der Termine länger dauerte als gedacht, musste ich mich richtig beeilen **und rennen**.
'Last Wednesday I had many appointments throughout the whole city. After one of the appointments took longer than expected, I had to hurry a lot (and run).'

The parts printed in bold in (34b) indicate the paraphrase of the gestures' meaning in (34a).

Additionally, in order to establish either a match or a mismatch, each experimental item was paired with two pictures: a picture matching the utterance and a picture where there was a mismatch between utterance and picture (factor Match: match vs. mismatch). The matching picture used for the experimental item in (34) can be found in Figure 3a and the mismatching picture is given in Figure 3b. Crucially, the (mis)match was always dependent on the manipulation of the factor Mode.

The study was designed along the lines of the first experiment reported in Ebert, Ebert & Hörnig (2020). The main difference is that the study reported here has a 3x2 design with the factors Mode and Match.

In order to distract participants from the research question, 24 unrelated filler items were included. The fillers were adapted from Ebert, Pirillo & Walter (2022). Each filler item consisted of two sentences. The first sentence introduced an event, and the second one further elaborated on this event. Crucially, the second sentence always contained a lexically ambiguous noun. This noun was then either accompanied by a matching or a mismatching gesture, i.e., a gesture that

(a) Picture for (35a) (mismatching)

(b) Picture for (35b) (matching)

Figure 4: Pictures used for the filler items given in (35)

either fit the salient interpretation of the noun in the given context or a gesture that contradicted the salient interpretation. The gesture was always aligned with the whole DP. Moreover, each filler was either paired with a picture matching or mismatching the event described in the verbal component of the utterance. The pictures were included to increase the similarity between the experimental materials and the fillers. An example of a filler item with a matching gesture is given in (35a) and an example of an item with a mismatching gesture is given in (35b).

(35) a. Silas geht jeden Tag auf den Fußballplatz. Den Ball bringt er von zu Hause mit.
'Silas goes to the soccer ground every day. He brings the ball with him from his home.'
+ Circular gesture tracing the ball's shape

b. Mariannes Hund hat Junge bekommen. Der ganze Wurf ist kerngesund.
'Marianne's dog got puppies yesterday. The whole litter (of puppies) is hale and healthy.'
+ Throwing gesture

The word *Ball* in German is ambiguous between ball in the sense of a sports device and in the sense of a festive dancing event, just as in English. The felicitous interpretation of ball in this context is the first one, i.e., ball as a sports device. Therefore, the gesture is matching in this example. The word *Wurf* is also ambiguous. It can either mean litter or toss. Since the sentence can only be felicitous under the first interpretation, the gesture does not match in this example. The pictures that accompanied the fillers in (35) are given in Figure 4. The picture for (35a) was a mismatching picture because it displays the ball in the sense of a dancing event, contrary to the felicitous interpretation of ball in this context. The picture for (35b), by contrast, matches the verbal part of the item because it shows a litter of puppies.

All items were videotaped and recorded in a professional studio in one session. The items were recorded by an acting student. The pictures used in the experimental study were taken either from shutterstock[5], a website where one can acquire the copyright for pictures, or from freeimages[6], pexels[7], and pixabay[8], which are websites where one can download license-free pictures.

[5] www.shutterstock.com
[6] www.freeimages.com
[7] www.pexels.com
[8] www.pixabay.com

4.1.3 Procedure

The procedure was similar to the first experiment reported in Ebert, Ebert & Hörnig (2020). Participants had to complete a web-based questionnaire where they had to rate how well a videotaped utterance matched a picture they had seen prior to the video. Completion took participants about 20 minutes.

Participants were made familiar with their task by an introductory text at the beginning of the questionnaire. They were instructed in this text to pay attention to the audio tape and the video tape when watching the videos. Additionally, the text informed them about their data protection, and they were also explicitly instructed that they could stop the questionnaire at any time without providing any explanation. After the introduction, there was a training session consisting of two trials.

The questionnaire was created using SoSci Survey (Leiner 2022), an online platform to design questionnaires which can be used free of charge for academic purposes. It was distributed via Prolific. Prolific's pre-filtering function was used in order to ensure that only native speakers of German participate in the study. Each list of the questionnaire was run independently, excluding for each list those participants who already completed the questionnaire.

The items were split up according to a Latin square design and evenly distributed onto six lists. All 24 fillers were included on each of the six lists. In addition, three questions about the videos were included in each questionnaire targeting participants' attention. They were, for example, asked about the color of the background in the videos. All participants answered the three questions correctly. The ordering of the items was randomized for each participant. Participants had to rate on a 7-point scale in the Likert response format how well the videotaped utterance matched the picture they had previously seen (1 = no match; 7 = perfect match).

4.2 Predictions

Recall the two research questions:

(36) 1. Do CVGs and OVGs contribute not-at-issue meaning by default?
2. Are there differences in the at-issue status of CVGs and OVGs?

As has been pointed out in Section 2 of the present paper, the negation test is a frequently used device to determine an expression's at-issue status. Example (37) illustrates this for the two types of viewpoint gesture targeted in this paper.

(37) A: Simon overslept today. He then had to <u>hurry</u> to get to work on time.
CVG: Speaker moves their arms and legs to demonstrate a running event.
OVG: Speaker uses two fingers of their right hand to depict a running event; each finger represents a leg.
B: #No, he did not run, he went by bicycle.
B': Hey, wait a minute! He had to hurry to get to work on time, but he did not run.
B": Yes, but actually he did not run, he went by bike.

Intuitively, it shows that CVGs and OVGs cannot be directly denied in discourse, hence causing the infelicity of B's utterance in the example. If a speaker wants to target a gesture's content, it seems much more plausible to use a discourse-interrupting element (cf. B') or to agree with the at-issue proposal and then deny the gesture's content (cf. B"). Based on this observation, it is hypothesized for Research Question 1 of the present paper that CVGs and OVGs both contribute not-at-issue meaning by default, in line with Ebert, Ebert & Hörnig's (2020) findings. It is thus

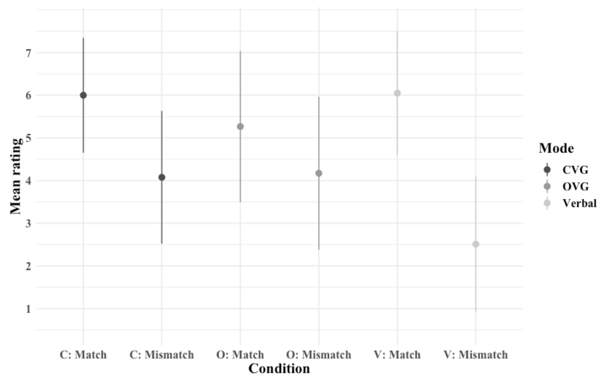


Figure 5: Mean values and SDs for each condition

predicted that the mismatch effect (i.e., the rating difference between the matching and the mismatching condition) is significantly higher for items in the condition Verbal compared with items in the CVG or OVG condition. In other words, an interaction between Match and Mode for the pairwise comparison between CVG and Verbal and OVG and Verbal is predicted.

It has been shown in Section 3.4 that CVGs are more informative than OVGs (Beattie & Shovelton 2002). Moreover, when producing a CVG, the whole body is usually involved. When producing an OVG, by contrast, only the hands and arms are involved (McNeill 1992). It is therefore hypothesized for Research Question 2 that CVGs are more at-issue than OVGs, although both cannot be directly denied in discourse. For Research Question 2, a higher mismatch effect for CVG items compared to OVG items is thus predicted. This means that an interaction of Match and Mode for the pairwise comparison of the conditions CVG and OVG is predicted.

The second prediction is troublesome under the standard assumption that at-issueness is conceptualized as a binary notion because the hypothesis that the two gesture types differ with respect to their at-issue status would automatically result in one gesture type being at-issue and one being not-at-issue by default. Since the test in (37) clearly suggests otherwise, a gradient conceptualization of at-issueness will be assumed here (cf. Barnes & Ebert 2023). Under this approach, the two gesture types can both be hypothesized to contribute not-at-issue meaning by default and still differ with respect to their at-issue status.

4.3 Results

The data presented in this section as well as Section 5 were analyzed using the R statistics software (R Core Team 2022). In order to test for significant effects, the results were analyzed using cumulative link mixed effects models with the function clmm() in the ordinal package (Christensen 2023). An ordinal mixed effects model was chosen to analyze the data instead of a linear mixed effects model mainly out of two reasons: i) linear mixed effects models require the data to be measured at the interval level, and ii) linear mixed effects models require the data to be normally distributed. Both is questionable for Likert scale data. Therefore, ordinal mixed effects models are more suitable to analyze this kind of data. For the statistic computation, the two factors and all the interactions between them were entered as fixed effects into the model using effect coding, i.e., the intercept represents the unweighted grand mean while the fixed effects compare the factor levels to each other. For Mode, the three-level factor, two pairwise contrasts were defined. Pairwise contrasts compare two levels of a factor with each other. Pairwise contrasts were calculated using the package emmeans (Lenth 2024). The stimuli, the raw data, and the analysis scripts for both experiments reported in this paper can be found

Table 1: Ordinal mixed effects model with Mode and Match as fixed effects and participants and items as random intercepts. Formula: ratingO $\sim$ Gesture * Perspective + (1|CASE) + (1|item) Significance codes: *** 0.001 | ** 0.01 | * 0.05 | . 0.1

	Estimate	Std. Error	z value	Pr(>|z|)
Match	-2.857	0.139	-20.60	<2e-16 ***
Mode – CVG vs. OVG	-0.423	0.136	-3.11	0.002 **
Mode – OVG vs. V	0.394	0.143	2.76	0.006 **
Match:Mode – CVG vs. OVG	1.176	0.273	4.30	1.7e-05 ***
Match:Mode – OVG vs. V	3.193	0.293	10.90	<2e-16 ***

here: `https://osf.io/2zvj7/`.

Figure 5 shows the mean ratings and the standard deviations (SD) for each condition. The means for CVG items were 6.00 (*SD* = 1.35) in the match condition and 4.08 (*SD* = 1.56) in the mismatch condition. For OVG items, the mean in the match condition was 5.27 (*SD* = 1.77) and in the mismatch condition it was 4.17 (*SD* = 1.79). Verbal items had a mean rating of 6.05 (SD = 1.45) in the match condition and a mean rating of 2.51 (*SD* = 1.58) in the mismatch condition.

The mixed effects model corresponding to the data in Figure 5 is given in Table 1. There is a main effect for the factor Match as well as an interaction between the factors Match and Mode. Pairwise contrasts show a significant difference between the comparison of CVG and OVG for Mode and Match (*z-ratio* = -6.99, $p < 0.0001$). Pairwise contrasts for CVG and Verbal as well as OVG and Verbal for Mode and Match also reached significance (CVG vs. Verbal: *z-ratio* = 4.3, $p = 0.0001$; OVG vs. Verbal: *z-ratio* = 10.9, $p < 0.0001$).

4.4 Discussion

The results show that the mismatch effect is significantly higher for Verbal items than for CVG items. The same can be observed for Verbal items as compared to OVG items. These findings were predicted for Research Question 1. Therefore, the results corroborate the hypothesis that CVGs and OVGs contribute not-at-issue meaning by default. The findings thus also replicate the results of the first experiment reported in Ebert, Ebert & Hörnig (2020) that co-speech gestures contribute not-at-issue meaning by default. The finding that the mismatch effect is significantly higher for CVG items as opposed to OVG items confirms the hypothesis that the two gesture types differ with respect to their at-issue status. In more detail, this means that CVGs are more at-issue than OVGs because the former gesture type induces a higher mismatch effect than the latter type.

The interpretation of the results has to be taken with some caution, however. Recall that participants were asked in each trial how well the utterance matched the picture. In other words, this question was the QUD of each trial. Now also note that—depending on the condition—the match or mismatch between utterance and picture only came about via the manipulation of the factor Mode. In other words, without the CVG, OVG, or verbalized version of the gesture provided by the factor Mode, each picture would always have matched the utterance. Therefore, the Mode manipulation and thus also the CVGs and OVGs in the respective condition provided QUD-relevant information. Now recall also the aforementioned core assumption made by Barnes & Ebert (2023) that expressions which are normally not-at-issue can shift toward the at-issue dimension in case they provide QUD-relevant information. Transferring this to the design of the study, this means that the QUD-relevance of CVGs and OVGs in the current experiment might have caused them to shift toward the at-issue dimension. This might have held especially strong for CVGs, as the mismatch effect observed for them was higher than that observed for OVGs. In order to control for this, a second experiment was conducted where the gestural information did not address the QUD.

5 Experiment 2

5.1 Method

5.1.1 Participants

45 participants were again recruited via Prolific. Their age span was 22–72 years (M = 40.43, SD = 13.04). They were native speakers of German who were naive with respect to the research question. For participation they received a compensation of £3.00.

5.1.2 Materials

The same recordings as for Experiment 1 were used. This time, however, the videotaped utterance was paired with a written response instead of a picture. An example of an experimental item is given in (38).

(38) a. Letzten Mittwoch hatte ich den ganzen Tag Termine überall in der Stadt. Nachdem einer der Termine länger dauerte als gedacht, musste ich mich <u>richtig beeilen</u>.
'Last Wednesday I had many appointments throughout the whole city. After one of the appointments took longer than expected, I had to <u>hurry a lot</u>.'
CVG: Speaker moves their arms and legs to demonstrate a running event.
OVG: Speaker uses two fingers of their right hand to depict a running event; each finger represents a leg.
b. Ja, stimmt, aber gerannt bist du ja nicht.
'Yes, true, but you didn't run.'

As an alternative to the gesture conditions in (38a), a verbalized version of the gestures was given again. The words printed in bold in (39) correspond to the verbalized contribution of the gestures in (38a).

(39) Letzten Mittwoch hatte ich den ganzen Tag Termine überall in der Stadt. Nachdem einer der Termine länger dauerte als gedacht, musste ich mich richtig beeilen **und rennen**.

The study was thus of a single factor design with the factor Mode (CVG vs. OVG vs. Verbal).

In addition, the same 24 fillers as in Experiment 1 were used, but also without a picture. In contrast to the experimental items, different response types were used. The denial and affirmation in (40) and (41), respectively, targeted the at-issue content of the filler.

(40) a. Mariannes Hund hat Junge bekommen. <u>Der ganze Wurf</u> ist kerngesund.
'Marianne's dog got puppies yesterday. <u>The whole litter (of puppies)</u> is hale and healthy.' + **Throwing gesture**
b. Nein, stimmt nicht, einem Welpen geht es leider nicht gut.
'No, that's not true, unfortunately one of the puppies is not well.'

(41) a. Franziska hat sich am Montag neue Schuhe gekauft. Gleich am nächsten Tag ist <u>der Absatz</u> abgebrochen.
'Franziska bought new shoes on Monday. On the next day, <u>the heel</u> broke off.' + **Showing the size of the heel with index finger and thumb**

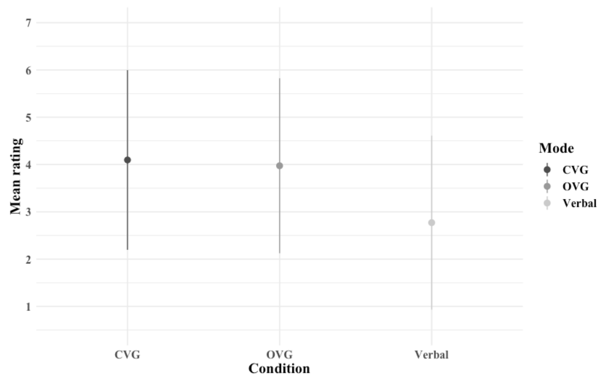


Figure 6: Mean values and SDs for each condition

b. Ja, stimmt, sie musste sie dann zurückgeben.
'Yes, true, she had to return them.'

Note that, as in Experiment 1, the gestures could either be matching or mismatching again. This was done in order to achieve bad as well as good ratings on the Likert scale.

5.1.3 Procedure

The procedure was the same as for Experiment 1, the main difference being that this time participants had to rate how acceptable they find the utterance in (38b) as a response to the videotaped utterance they had seen (cf. (38a)). This was done in order to ensure that the task, which participants most likely reconstructed as the current QUD in Experiment 1, did not directly target the gesture's meaning contribution this time because this might shift them toward the at-issue dimension (Barnes & Ebert 2023). They had to rate the sentences on a scale from 1 to 7 (1 = completely unacceptable; 7 = completely acceptable).

5.2 Predictions

The response used for the experimental items can only be used felicitously to target not-at-issue content (Tonhauser 2012). Therefore, based on the hypothesis and also the results of Experiment 1, first and foremost rating differences between the condition Verbal and the two gesture conditions are predicted. Moreover, since the results of Experiment 1 suggest that CVGs, although not-at-issue by default, are more at-issue than OVGs. Therefore, based on the hypothesis and the results of Experiment 1, rating differences are primarily predicted between the Verbal condition and the two gesture conditions. Moreover, since the results of Experiment 1 suggest that CVGs, although not at issue by default, are more at issue than OVGs, this should also be reflected in the ratings. Specifically, items in the CVG condition are predicted to be rated worse than items in the OVG condition. Thus, a main effect of Mode is predicted.

5.3 Results

The mean values and SDs are shown in Figure 6. In general, no big rating differences between the condition CVG (M = 4.1, SD = 1.9) and the condition OVG (M = 3.97, SD = 1.85) can be observed. What can be seen, however, is that for the condition Verbal the ratings were worse (M = 2.77, SD = 1.84) than for both gesture conditions. In order to test for the hypothesized main effect, an ordinal mixed effects model was fitted onto the data. The model output is given

Table 2: Ordinal mixed effects model with Mode fixed effect and participants and items as random intercepts. Formula: ratingO $\sim$ Gesture * Perspective + (1|CASE) + (1|item)
Signif. codes: *** 0.001 | ** 0.01 | * 0.05 | . 0.1

	Estimate	Std. Error	z value	Pr(>|z|)
Mode – CVG vs. OVG	-0.091	0.155	-0.59	0.56
Mode – OVG vs. Verbal	1.59	0.164	9.696	<2e-16 ***

in Table 2. The pairwise contrast between CVG and Verbal (*z-ratio* = 9.696, $p < 0.0001$) as well as the contrast between OVG and Verbal reached significance (*z-ratio* = 9.063, $p < 0.0001$). However, no difference between the two gesture conditions—CVG and OVG—can be observed (*z-ratio* = 0.059, $p = 0.826$).

5.4 Discussion

The results suggest that, in line with the findings of Experiment 1, that CVGs and OVGs both contribute not-at-issue meaning by default. The findings thus also replicate Ebert, Ebert & Hörnig's (2020) results, thereby providing evidence in favor of accounts of gesture semantics that posit that co-speech gestures are not-at-issue (Ebert & Ebert 2014; Schlenker 2018a). However, the results are also somewhat puzzling. Against the findings of Experiment 1, no significant rating difference between the two gesture conditions could be observed, which was predicted under the assumption that CVGs are more at-issue than OVGs.

6 General discussion and conclusion

Most semantic accounts on the meaning contribution of co-speech gestures posit that co-speech gestures contribute not-at-issue meaning by default (Ebert & Ebert 2014; Schlenker 2018a: cf. Section 3.1 of the present paper). This has been experimentally validated for co-speech gestures giving shape information of an object (Ebert, Ebert & Hörnig 2020), that means for gestures which do not encode viewpoint. Moreover, a second experiment reported in Ebert, Ebert & Hörnig (2020) investigated the claim that demonstratives, such as the German demonstrative *SO* ('so/like this/such'), function as so-called dimension shifters, meaning that they shift the gesture's meaning contribution to the at-issue dimension (Ebert & Ebert 2014). The results of this experiment partly confirm this claim: while the demonstrative indeed shifted the gesture's information status more toward the at-issue dimension, it was arguably still less at-issue than asserted material. This and similar data from sentence-medial adverbial ideophones in German (Barnes et al. 2022) lead Barnes & Ebert (2023) to the claim that at-issueness is better captured as a gradient notion (cf. Section 3.2) than a binary one (cf. Section 2 of the present paper). Barnes & Ebert (2023) assume that every expression comes with an inherent at-issue status based on whether it is normally used to address the QUD or not. If an expression that typically does not address the QUD exceptionally does so, it shifts toward the at-issue dimension.Their account thus also predicts the shifting behavior of co-speech gestures aligned with German *SO*: the combination of a demonstrative with a gesture simply is assigned a higher inherent at-issue status as a co-speech gesture aligned with other lexical expressions. At the same time, this value is lower than that of asserted material.

In this paper, the results of two experimental studies were reported investigating the at-issue status of viewpoint gestures, more specifically of CVGs and OVGs. The studies were conducted as it was unclear whether Ebert, Ebert & Hörnig's (2020) findings could be straightforwardly applied to viewpoint gestures, especially CVGs, since CVGs are normally produced with the whole body (McNeill 1992, see Section 3.4 of the present paper) and thus differ in size not only from OVGs, but also from the gestures used in the study of Ebert, Ebert & Hörnig (2020). Moreover, as has been shown in Section 3.4, CVGs and OVG differ in various aspects: for example, CVGs are more informative (Beattie & Shovelton 2002). Based on these differences between the two types of viewpoint gestures, it was hypothesized that while both gesture types contribute not-at-issue meaning by default, CVGs are more at-issue than OVGs due to the aforementioned differences between them. A gradient approach to at-issueness was thus adopted in this paper.

The first experiment, reported in Section 4, adapted the design of Ebert, Ebert & Hörnig (2020). Based on the hypothesis that CVGs are more at-issue than OVGs, it was hypothesized that the mismatch effect should be higher for CVGs compared to OVGs. The mismatch effect in the verbal control condition was predicted to be highest. The results confirmed this hypothesis. In the discussion, it was noted, however, that the findings should be taken with a grain of salt because the question participants were asked before giving their rating for each item explicitly targeted the content of the gesture. It was thus possible that they reconstructed this question as a QUD to which the gestures gave at least a partial answer. Under the gradient account to at-issueness proposed by Barnes & Ebert (2023), this might have caused the CVG items to shift toward the at-issue dimension. In order to address this matter, the second experiment, which was reported in Section 5, was conducted. In the experiment, one of the diagnostics for not-at-issue content proposed in Tonhauser (2012) was used, and the question participants were asked did not explicitly target the gesture's content. The results could only partially confirm the results obtained from Experiment 1: while they validated that both gesture types contribute not-at-issue meaning by default, no rating differences between CVGs and OVGs could be observed.

Taking together the results of both studies, they can be interpreted as follows: in cases where they do not addresss the QUD, CVGs and OVGs are equally not-at-issue, in line with Ebert, Ebert & Hörnig (2020). However, as soon as they address the QUD, CVGs can shift more toward the at-issue dimension than OVGs can. The results still show, however, that CVGs remain in the not-at-issue dimension, as they did not pattern with the verbal control condition in Experiment 1. This constitutes evidence in favor of the gradient approach to at-issueness advocated by Barnes & Ebert (2023). A question that remains open for now is why CVGs can shift more toward the at-issue dimension than OVGs can. A plausible explanation for this is as follows: CVGs are larger in size than OVGs. In addition, they have been shown to be more informative (Beattie & Shovelton 2002). Their higher informativity is potentially related to their larger size, and therefore, speakers can encode more information in a CVG than in an OVG. This, in turn, results in listeners being able to also extract more information from CVGs compared to OVGs. Thus, they can use more information to give a richer (partial) answer to the QUD. Future research should investigate whether this is on the right track. In general, the interplay of informativity and at-issue status constitutes an interesting avenue for future research.

An alternative explanation of the data could also be methodological in nature. The mismatch effect adapted from Ebert et al. (2020) and used in Experiment 1 works well in experimental settings for testing the at-issue status of iconic enrichments in particular. However, it relies on a strong correlation between acceptability and truth (cf. Koev 2023 for a more in-depth discussion). The diagnostic used in Experiment 2, introduced by Tonhauser (2012), has the drawback that it was originally designed only for spoken and written content, not for gestures. To test the at-issue status of a gesture using this method, it must be translated into the spoken or written modality for denial, which could itself be a confounding factor in the data. It is therefore also possible that the differing results reported in the studies presented in this paper are attributable to differences in the at-issueness tests employed. This should also be addressed in future research.

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